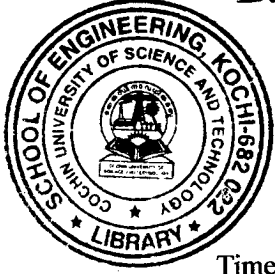


**B.Tech. Degree II Semester Regular and Supplementary/
I Semester Supplementary Examination May 2017**

CS/EC/IT AS 15-1201 B & CE/EE/ME/SE AS 15-1101 A

CALCULUS
(2015 Scheme)



Time: 3 Hours

Maximum Marks: 60

PART A(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Solve $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$.
- (b) Solve $\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$.
- (c) Solve $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 3x = \sin t$.
- (d) Test the convergence of the series $\sum \frac{1}{\sqrt{n} + \sqrt{n+1}}$.
- (e) Find the Maclaurin series expansion of $\tan^{-1} x$.
- (f) If $x^y + y^x = c$, find $\frac{dy}{dx}$.
- (g) Show that $U = e^x \sin y$, $V = e^x \cos y$ are functionally independent.
- (h) Find the perimeter of the cardioid $r = a(1 - \cos \theta)$.
- (i) Find the area enclosed by the curve given by the equation $x = a \cos^3 \theta$, $y = b \sin^3 \theta$.
- (j) Evaluate $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz$.

PART B

(4 × 10 = 40)

- II. (a) Solve $\frac{d^4x}{dt^4} + 4x = 0$.
- (b) Solve $\frac{d^2y}{dx^2} - \frac{2dy}{dx} + y = xe^x \sin x$.
- (c) Solve $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 20y = (x+1)^2$.

OR

(P.T.O.)

- III. (a) Solve the simultaneous equation.

$$\frac{dx}{dt} - x + \frac{dy}{dt} = 2t + 1$$

$$2\frac{dx}{dt} + x + \frac{2dy}{dt} = t.$$

- (b) A condenser of capacity C discharged through inductance L and resistance R in series and the charge q at time t satisfies the equation

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0, \text{ where } L = 1 \text{ henry, } R = 400 \text{ ohms, } C = 16 \times 10^{-6} \text{ farads.}$$

Find q in terms of t .

- IV. (a) Find the nature of the series.

$$\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \dots$$

- (b) State Raabe's test and test the convergence of

$$1 + \frac{1}{2}x + \frac{1.2}{2.5}x^2 + \frac{1.2.3}{2.5.8}x^3 + \dots$$

OR

- V. (a) Define conditional convergence and absolute convergence of series with examples.

- (b) If $y = e^{a \sin^{-1} x}$, prove that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + a^2)y_n = 0$.

Hence find the value of y_n when $x = 0$.

- VI. (a) Show that if $u = \sin^{-1} \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0$.

- (b) If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

- (c) The diameter and height of a right circular cylinder are measured to be 5 and 8 inches respectively, if each of these dimensions are in error 0.1 inch. Find the relative percentage error in volume of the cylinder.

OR

- VII. (a) If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$. Show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta.$$

- (b) A rectangular box open at the top is to have a volume 32 c.c. Find the dimensions of the box requiring least material for its construction.

- VIII. (a) Find the area of the surface of revolution formed by revolving the loop of the curve $9ay^2 = x(3a-x)^2$ about X -axis

- (b) Evaluate the integral by changing the order of integration $\int_0^a \int_{x/a}^{2a-x} xy dy dx$.

OR

- IX. (a) Calculate $\iint r^3 dr d\theta$ over the area included between the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$.

- (b) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$.